

Step by step through running **SQL Server on Amazon EC2** (Infrastructure as a Service), including **architecture, setup, HA, cost model, live examples, and real-world use cases.**

1. What is SQL Server on Amazon EC2?

Amazon EC2 (Elastic Compute Cloud) lets you run **virtual machines** (called “instances”) in AWS.

When you deploy **SQL Server on EC2**, you’re installing SQL Server on a **Windows (or Linux) virtual machine** — just like you would on-premises.

You have:

- **Full control** over the OS and SQL Server configuration.
- **Direct access** via Remote Desktop (RDP) or SSH.
- **Responsibility** for patching, backups, tuning, HA, and security.

👉 In short:

You manage **everything inside the instance**, while AWS provides **the compute, network, and storage infrastructure.**

2. EC2 vs RDS vs Aurora — Quick Comparison

Feature	EC2 (IaaS)	RDS (PaaS)	Aurora (Cloud-native)
Control	Full OS + SQL access	Limited (DB only)	Fully managed
Patching	Manual	Automatic	Automatic
Backups	Manual/custom	Automated	Automated
HA/DR	Custom setup	Built-in Multi-AZ	Built-in
Best for	Legacy apps, custom configs	General web apps	Scalable cloud-native
Pricing	Compute + storage	Instance + managed services	Cluster-based

✅ Choose EC2 if:

- You need **full control** (OS-level access, custom configurations).
- Your SQL setup involves **custom agents, linked servers, SSIS, CLR, etc.**
- You’re migrating an existing on-prem SQL Server “as-is”.

3. Architecture Overview

A typical **SQL Server on EC2** setup includes:

1. **Amazon EC2 Instance**
 - Windows Server 2022 or 2019.
 - SQL Server installed manually or via AWS Marketplace AMI.
2. **Storage**
 - EBS Volumes (separate for OS, Data, Logs, TempDB).
 - Optionally attach additional disks for performance.
3. **Networking**
 - Inside a VPC (Virtual Private Cloud).
 - Access controlled by Security Groups and NACLs.
4. **Backup**
 - SQL native backups → Amazon S3.
 - Or automate via AWS Backup service.

5. High Availability

- Use Always On Availability Groups or Database Mirroring.
- Secondary EC2 instance in another Availability Zone.

6. Monitoring

- Use CloudWatch, CloudTrail, and SQL native monitoring tools.



4. Deployment Methods

Option 1: From AWS Marketplace

- Launch **pre-built AMI** with SQL Server.
- Editions: Express, Web, Standard, Enterprise.
- License models:
 - **License Included** (PAYG)
 - **Bring Your Own License** (BYOL)

Option 2: Manual Installation

- Launch Windows EC2 instance.
- Install SQL Server from your own ISO or license key.
- Configure storage and security.



5. Live Example: Deploy SQL Server EC2 Instance

Let's simulate a **real-world setup** 📌

Scenario:

You're deploying **SQL Server 2022 Standard** for an ERP application.

Steps:

1. Create the Instance

- Go to AWS Console → EC2 → Launch Instance
- Choose AMI: *Microsoft Windows Server 2022 with SQL Server Standard 2022*
- Choose instance type: m6i.xlarge (4 vCPU, 16 GB RAM)
- Storage:
 - C: (OS) → 100 GB gp3
 - D: (Data) → 200 GB gp3
 - E: (Logs) → 100 GB gp3
 - F: (TempDB) → 50 GB gp3

2. Networking

- Place in your application **VPC** and **private subnet**.
- Open SQL port (1433) for app servers only.

3. Connect

- RDP into the Windows instance.
- Open SQL Server Management Studio (SSMS).

4. Create a Database

```
CREATE DATABASE ERPDB;
```

```
USE ERPDB;
```

```
CREATE TABLE Customers (
```

```
    ID INT IDENTITY(1,1) PRIMARY KEY,
```

```
    Name NVARCHAR(100),
```

```
    City NVARCHAR(50)
```

```
);
```

```
INSERT INTO Customers (Name, City) VALUES ('Alice', 'New York'), ('Bob', 'Chicago');
```

```
SELECT * FROM Customers;
```

5. Backups

- Create a folder D:\Backups.
- Schedule SQL Agent Job:
- `BACKUP DATABASE ERPDB TO DISK = 'D:\Backups\ERPDB_FULL.bak' WITH INIT;`
- Upload to S3:
- `aws s3 cp D:\Backups\ERPDB_FULL.bak s3://my-sql-backups/`

6. Automation (Optional)

- Use **AWS Systems Manager (SSM)** to schedule patching or monitoring scripts.

✅ You now have a full SQL Server environment running on EC2, under your control.

6. High Availability & Disaster Recovery (HA/DR)

Several options are available:

HA/DR Method	Description	Use Case
Always On Availability Groups	Real-time replication between primary and secondary EC2 instances in different AZs.	Mission-critical workloads
Failover Cluster Instance (FCI)	Shared storage cluster; requires FSx for Windows File Server.	Enterprise environments
Log Shipping	Manual setup for periodic DR replication.	Simple, cost-effective
AWS Backup or S3 Snapshots	Scheduled backups stored in S3.	Low-cost protection

7. Security Best Practices

Layer	Recommendation
Network	Place EC2 in private subnet; control inbound rules with Security Groups.
Authentication	Integrate with Active Directory or use SQL authentication.
Encryption	Enable Transparent Data Encryption (TDE) .
Storage	Encrypt EBS volumes using KMS .
Backups	Store encrypted backups in S3 with access policies.
Access	Use AWS IAM roles for S3 and Systems Manager operations.

8. Monitoring & Maintenance

Tool	Purpose
Amazon CloudWatch	Monitors CPU, disk IOPS, memory, network throughput
SQL Profiler / Extended Events	Query-level performance tuning
CloudTrail	Logs API calls and access events
AWS Systems Manager	Patching and command automation
AWS Backup	Scheduled, centralized backup of SQL EC2

💰 9. Cost Breakdown

Main cost components:

Component	Description
EC2 Instance Type	e.g., m6i.xlarge (4 vCPU, 16 GB RAM)
Storage (EBS)	SSD gp3 or io2 disks for data, logs
SQL License	Included (PAYG) or BYOL
Backup Storage	S3 costs for backups
Data Transfer	Intra-VPC is free; external transfer costs extra

Example Monthly Estimate:

- Instance: m6i.xlarge → ~\$160
 - SQL Standard License → ~\$200
 - 400 GB gp3 storage → ~\$40
- ✅ ≈ \$400/month total

🧠 10. Real-World Usage Scenarios

Scenario	Why EC2 (SQL Server)?
Legacy ERP / CRM apps	Full OS-level control and SQL Agent jobs.
Custom BI / ETL pipelines	Requires SSIS, SSAS, or CLR integration.
High-performance on-prem migration	Lift-and-shift existing SQL setups.
Hybrid environments	Connects via VPN/Direct Connect to on-prem AD.
SQL Server with non-Microsoft components	Install antivirus, monitoring agents, or scripts freely.
Multi-instance setup	Host multiple SQL instances on same EC2 (not possible in RDS).

⚡ 11. Performance Best Practices

1. Separate disks:
 - Data → 1 EBS volume
 - Logs → another
 - TempDB → high IOPS SSD
2. Use **Provisioned IOPS (io2)** for mission-critical databases.
3. Enable **SQL Instant File Initialization**.
4. Configure **NUMA & MAXDOP** properly.
5. Monitor with **Performance Insights** (via CloudWatch or SSMS).
6. Periodically resize instance type for growth.

12. Summary

Feature	Description
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Type	IaaS (self-managed)
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Control	Full OS + SQL customization
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Patching	Manual or scripted
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Backup	User-managed (SQL or AWS Backup)
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HA/DR	Always On, FCI, Log Shipping
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Security	Full control with KMS, AD, IAM
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Cost	Instance + storage + license
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Best Use	Legacy migration, full control, or custom SQL workloads
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